EXPERIMENT 7

COLUMNAR TRANSPOSITION CIPHER

AIM

To implement the Columnar Transposition Cipher using Python.

ALGORITHM

1. Read the plaintext and keyword.

2. Arrange the plaintext row-wise in a matrix.

3. Number the columns according to the alphabetical order of the keyword.

4. Read the columns in sorted-key order.

5. Display the encrypted text.

PYTHON CODE

```
import math


text = input("Enter Plain Text: ").replace(" ", "").upper()

key = input("Enter Key: ").upper()


col = len(key)

row = math.ceil(len(text) / col)


fill = row * col - len(text)

text += "X" * fill


matrix = []


k = 0

for i in range(row):

    matrix.append(list(text[k:k + col]))

    k += col
```

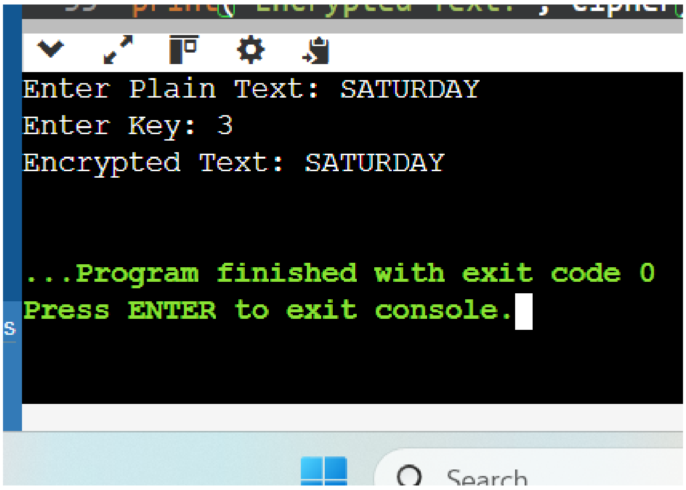
```
order = sorted(list(enumerate(key)), key=lambda x: x[1])

cipher = ""

for index, _ in order:
    for r in matrix:
        cipher += r[index]

print("Encrypted Text:", cipher)
```

OUTPUT



RESULT

Thus the Columnar Transposition Cipher was implemented successfully using Python.